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VEHICLE COMMUNICATIONS

Abstract

A vehicle may detect an entity within a specified distance of a vehicle, output a first audio outside of the vehicle in response to detecting the entity, classify the entity based on data collected about the entity after outputting the first audio, and output a second audio based on classifying the entity.

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Background/Summary

BACKGROUND

[0001] Vehicles typically include sensors for collecting data about the vehicle and/or an

environment around the vehicle. In some examples, sensor data can be used by vehicle systems to actuate vehicle components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1 is a block diagram of an example vehicle.

[0003] FIG. 2 illustrates an example vehicle detecting an entity.

[0004] FIG. 3 is a process flow diagram illustrating an example process for actuating vehicle components based on entities proximate to the vehicle.

DETAILED DESCRIPTION

Introduction

[0005] Described herein are techniques for detecting and interacting with entities within a specified distance of a vehicle. A vehicle computer may classify an entity based on data collected by vehicle sensors. The vehicle may then actuate components based on the classification of the entity. For example, the vehicle may output audio, unlock doors, send a message to a remote computer, etc.

[0006] Accordingly, included in the present disclosure is a system comprising a computer that includes a processor and a memory, the memory storing instructions executable by the processor to: detect an entity within a specified distance of a vehicle, output a first audio outside of the vehicle in response to detecting the entity, classify the entity based on data collected about the entity after outputting the first audio, and output a second audio based on classifying the entity.

[0007] A vehicle feature may be activated based on classifying the entity.

[0008] The detection of the entity within the specified distance of the vehicle may be based on image data.

[0009] The second audio may be responsive to a communication by the entity.

[0010] The second audio may be selected from a stored plurality of phrases.

[0011] The entity may be identified based on stored data collected from the entity prior to outputting the first audio.

[0012] The entity may be classified based on a plurality of images collected at different times.

[0013] The entity may be classified based on a tone of the entity.

[0014] The classification of the entity may be one of threatening or nonthreatening.

[0015] The first audio may be output in a first language and the second audio may be output in a second language based on detecting the second language in an entity audio.

[0016] A first entity and a second entity may be detected within the specified distance of the vehicle, the first entity being prioritized based on a risk score.

[0017] A method comprises: detecting an entity within a specified distance of the vehicle, outputting a first audio outside of the vehicle in response to detecting the entity, classifying the entity based on data collected about the entity after outputting the first audio, and outputting the second audio based on classifying the entity.

[0018] A vehicle feature may be activated based on classifying the entity.

[0019] The entity may be detected within the specified distance of the vehicle based on image data.

[0020] The second audio may be responsive to a communication by the entity.

[0021] The second audio may be selected from a stored plurality of phrases.

[0022] The entity may be identified based on stored data collected from the entity prior to outputting the first audio.

[0023] The entity may be classified based on a tone of the entity.

[0024] The classification of the entity may be one of threatening or nonthreatening.

[0025] The first audio may be output in a first language and the second audio may be output in a second language based on detecting the second language in an entity audio.

Exemplary System Elements

[0026] FIG. 1 is a block diagram of a vehicle system **100** for providing digital images for vehicle operation. The vehicle **102** includes a computer **104** having a memory that includes instructions executable by the computer **104** to carry out processes and operations, including as described herein. The computer **104** may be communicatively coupled via a communication network, such as a vehicle network **114**, with sensors **106**, components **108**, a human machine interface (HMI) **110** and a communication module **112** in the vehicle **102**. The vehicle **102** may be any passenger or commercial automobile such as a car, a truck, a sport utility vehicle, a crossover, a van, a minivan, a taxi, a bus, ICE, BEV, Hybrid, etc.

[0027] As mentioned above, the vehicle computer **104** includes a processor and a memory. The memory includes one or more forms of computer readable media, and stores instructions executable by the vehicle computer **104** for performing various operations, including as disclosed herein. For example, a vehicle computer **104** can be a generic computer with a processor and memory as described above and/or may include an electronic control unit ECU or controller for a specific function or set of functions, and/or a dedicated electronic circuit including an ASIC (application specific integrated circuit) that is manufactured for a particular operation (e.g., an ASIC for processing sensor data and/or communicating the sensor data). In another example, a vehicle computer **104** may include an FPGA (Field-Programmable Gate Array) which is an integrated circuit manufactured to be configurable by a user. Typically, a hardware description language such as VHDL (Very High Speed Integrated Circuit Hardware Description Language) is used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured based on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming (e.g. stored in a memory electrically connected to the FPGA circuit). In some examples, a combination of processor(s), ASIC(s), and/or FPGA circuits may be included in a computer **104**.

[0028] The memory can be of any type (e.g., hard disk drives, solid state drives, servers, or any volatile or non-volatile media). The memory can store the collected data sent from the sensors **106**. The memory can be a separate device from the computer **104**, and the computer **104** can retrieve information stored by the memory via a network **114** in the vehicle **102** (e.g., over a CAN bus, a wireless network, etc.). Alternatively or additionally, the memory can be part of the computer **104** (e.g., as a memory of the computer **104**).

[0029] The vehicle computer **104** may include programming to operate one or more of vehicle brakes, propulsion (e.g., control of acceleration in the vehicle **102** by controlling one or more of an internal combustion engine, electric motor, hybrid engine, etc.) steering, climate control, interior and/or exterior lights, etc., as well as to determine whether and when the computer **104**, as opposed to a human operator, is to control such operations.

[0030] The computer **104** may include or be communicatively coupled to (e.g., via a vehicle network **114** such as a communications bus as described further below) more than one processor, (e.g., included in components **108** such as sensors **106**, electronic control units (ECUs) or the like included in the vehicle **102** for monitoring and/or controlling various vehicle components **108**, e.g., a powertrain controller, a brake controller, a steering controller, etc.). The computer **104** is generally arranged for communications on a vehicle communication network **114** that can include a bus in the vehicle **102** such as a controller area network CAN or the like, and/or other wired and/or wireless mechanisms. Alternatively or additionally, in cases where the computer **104** actually comprises a plurality of devices, the vehicle communication network **114** may be used for communications between devices represented as the computer **104** in this disclosure. Further, as mentioned below, various controllers and/or sensors **106** may provide data to the computer **104** via the vehicle communication network **114**.

[0031] Via the vehicle network **114**, the vehicle computer **104** may transmit messages to various

devices in the vehicle **102** and/or receive messages (e.g., CAN messages) from the various devices (e.g., sensors **106**, ECUs, etc.). Alternatively, or additionally, in cases where the vehicle computer **104** actually comprises a plurality of devices, the vehicle communication network **114** may be used for communications between devices represented as the vehicle computer **104** in this disclosure. Further, as mentioned below, various controllers and/or sensors **106** may provide data to the vehicle computer **104** via the vehicle communication network **114**.

[0032] The vehicle **102** typically includes a variety of sensors **106**. A sensor **106** is a device that can obtain one or more measurements of one or more physical phenomena. Some sensors **106** detect internal states of the vehicle **102**, for example, wheel speed, wheel orientation, and engine and transmission variables. Some sensors **106** detect the position or orientation of the vehicle **102**, for example, global positioning system GPS sensors. Some sensors **106** detect objects, for example, radar sensors, scanning laser range finders, light detection and ranging LIDAR devices, and image processing sensors such as cameras. Further sensors **106** detect sounds, for example dynamic or condenser microphones, piezoelectric transducers, ultrasonic sensors, acoustic emission sensors, etc. Such sensors for detecting sound are herein referred to as audio sensors. Audio sensors detect sound waves by measuring vibrations and converting them to electrical signals.

[0033] The vehicle **102** includes a variety of audio sensors (e.g., microphones, that can detect sound internal to the vehicle **102** and/or external to the vehicle **102**). An audio sensor may include or be connected to an analog-to-digital (A/D) converter to facilitate the conversion of sound waves to electrical signals. As an example, the A/D converter may utilize sampling to convert sound to electrical signals. An audio sensor and/or A/D converter may be in communication with the vehicle computer **104** such that the electrical signals may be transmitted to the vehicle computer **104**. One or more audio sensors may be installed or deployed in or on any suitable part of the vehicle **102** to provide detection of audio phenomena as described herein. For example, audio sensors for capturing sound emitted from or within the interior of the vehicle **102** may be provided in a vehicle dash panel, interior sides of a vehicle **102**, the interior of the vehicle roof, etc. As another example, audio sensors for capturing sound emitted exterior to the vehicle **102** may be supported by the exterior sides of the vehicle **102**, the exterior of the vehicle roof, in or on vehicle bumpers or fascia, the vehicle hood, etc. Multiple audio sensors may be installed at different positions in or on the vehicle **102** in order to provide sound capture and/or distinguish between multiple sound sources as described herein. The audio sensors may be any suitable type for detecting sound as described herein.

[0034] The vehicle **102** may include one or more sound output devices **116** (i.e., speakers). A speaker is a device that can output sound. As an example, a speaker can include a digital-to-analog (D/A) converter that converts electrical signals to vibrations to generate sound at a specified frequency. As an example, the D/A converter may input the electrical signal to a piecewise constant function. A speaker may be in communication with the vehicle computer **104** so as to receive electrical signals to be transmitted by the speaker in the form of sound. The speaker may be any suitable type of speaker to output sound internal to the vehicle **102** and/or external to the vehicle **102**. A speaker may also operate as a microphone. That is, a device may output audio as well as detect audio. In this document, sound output devices **116** and sensors **106** are referred to as separate elements, though it is to be understood that the operations of a speaker and a microphone elements could be performed by a single sound output device **116**. The speaker may be deployed or installed in or on the vehicle **102** (e.g., in or on a surface of the vehicle **102**). As an example, the speaker may be supported by the hood of the vehicle **102** so that output sound is likely to be detectable by listeners positioned in front of the vehicle **102**.

[0035] FIG. 2 illustrates an example scenario in which an entity **118** could be detected by a vehicle computer **104** from vehicle sensor data. The entity **118** may be any object or event detectable by the vehicle sensors **106**. In various examples described herein, the entity **118** is a person, though as further non-limiting examples the entity **118** could be a second vehicle, a fire, etc.

[0036] As described above the entity **118** may be any object detectable from data received from vehicle sensors **106**. A vehicle computer **104** could use any suitable object recognition algorithm or technique to detect and/or classify an object. For example, a neural network could be trained using data collected about objects to classify objects (e.g., to distinguish persons from other objects based on camera sensor data).

[0037] As another example of an entity **118**, the entity **118** may be an event such as a fire, an altercation between two or more people, etc. That is, the entity **118** may include detected happenings that are not objects. Any suitable technique to recognize and/or classify objects could be used to detect events (e.g., a neural network could be trained for object recognition or classification). Further, a neural network or the like could process sensor data in addition to or as an alternative to image data. Thus, in an example where the entity **118** is a fire, the vehicle may include sensors **106** which are heat sensors or thermal cameras. Data collected by such sensors could include an image, and/or could include a temperature of the entity **118**. If the temperature exceeds, for example, 160 degrees Fahrenheit, and/or based on image data, a computer **104** could classify the entity **118** as a fire. As another example, if the entity is an altercation, a computer **104** could input image data and/or audio data to one or more neural networks trained to classify altercations.

[0038] The vehicle computer may identify an entity. Identifying the entity herein means determining who the entity is (i.e., determining that the entity is a particular individual). For example, an entity could be classified as a person, and then identified as “Jane Doe.” The vehicle computer may identify an entity using any suitable recognition technique based on stored or collected data. As an example, if the vehicle computer has stored data about the entity such as an image of the face of the entity, the vehicle may compare the stored image to a newly collected image of the face of an entity. If the computer, by image recognition or any other suitable technique, determines that the two images match, the computer has identified the entity as the same entity from whom the stored data was collected.

[0039] As seen in FIG. 2, a sensor **106** may be positioned in or on the vehicle **102** to provide a field of view **120** extending relative to the vehicle body. The sensor **106** may be any type suitable to detect an entity **118** such as a camera, lidar, radar, etc. In the present example, the sensor **106** is a camera. The field of view **120** can be determined by specifications of the camera (e.g., a viewing angle providing by the camera) and pose or orientation of the camera (e.g., a pitch, roll, and yaw of an axis of the camera lens relative to a horizontal plane of the vehicle **102**). The field of view **120** is referred to as fixed or static when the camera is installed such that its pose cannot be changed. Additionally or alternatively, the camera may be moveably supported by the vehicle **102** and actuatable by the vehicle computer **104** such that the camera pose and/or a focus or orientation of a lens can be changed, and field of view **120** thereby dynamically adjusted.

[0040] A camera can generate image data of its field of view **120**. Image data from a camera typically includes a sequence of image frames from the field of view **120** (i.e., image data may be video data). The image frames can be input to the vehicle computer **104** via the vehicle network **114**. The vehicle computer **104** can analyze the image frames (e.g., can utilize suitable image recognition techniques) and may make a determination of whether the entity **118** is within a specified distance **122** of the vehicle **102** as described further below.

[0041] The vehicle computer **104** may be programmed to detect an entity **118** within a specified distance **122** of the vehicle **102**. For example, the vehicle computer **104** may detect the entity **118** and determine that the entity **118** is within the specified distance **122** of the vehicle **102** (e.g., measured as a radius around a point defined in or on the vehicle **102**, as a distance from an edge of a vehicle **102** as defined by a vehicle body, or based on any other suitable reference) based on collected data about the entity **118**. The vehicle computer **104** can make a determination of an entity **118** being within a specified distance **122** of the vehicle **102** when it determines that the entity **118**, or at least some portion of the entity **118**, is detected within a distance of the vehicle **102**

that is less than the specified distance **122**. The detection of the entity **118** within the specified distance **122** of the vehicle **102** may be based on data captured by the sensors **106**. As an example, the data may be image data. The distance may be measured from any selected point on the vehicle **102**. As an example, the distance may be measured from the sensor **106** which collected the entity data. As a further example, the distance may be measured from a surface of the vehicle **102** closest to the entity **118**.

[0042] The specified distance **122** may be any suitable distance determined by the vehicle computer **104** and or a vehicle operator. For example, the vehicle computer **104** may determine a specified distance **122** based on the vehicle **102** entering a geographic location (i.e., the vehicle computer **104** could store specified distances **122** for respective geographic locations, and could retrieve a specified distance **122** for a location from memory based on data from a location sensor, e.g., a GPS sensor). A specified distance **122** (i.e., a distance from a vehicle **102** within which detection of an entity **118** triggers classification and interaction with the entity **118**) may be determined based on characteristics or attributes of a geographic location. For example, a higher specified distance **122** of 20 meters could be selected when in geographic locations where vehicles are less likely to be parked closely to one another (e.g., suburban residential neighborhoods or rural locations). On the other hand, a lower specified distance **122** of 10 meters could be selected where vehicles are likely to be parked more closely to one another (e.g., an urban location, a parking garage or parking lot, etc.). A lower specified distance **122** may increase a frequency with which entities are classified and audio is output, and vice versa. Alternatively or additionally, a specified distance **122** may be selected by a vehicle operator. As an example, if the vehicle operator desires the vehicle **102** to only interact with entities which are close to the vehicle **102**, the vehicle operator could set the specified distance **122** to 1 meter. The vehicle operator may select the specified distance **122** by any suitable means such as by providing input via a vehicle HMI **110**.

[0043] The specified distance may be measured from the vehicle **102** while the vehicle **102** is stationary or while the vehicle is in motion. The entity **118** may be moving relative to the vehicle **102** while the vehicle is stationary or in motion and still be detected by the computer. As an example, the vehicle **102** may be backing down a driveway and an entity **118** may be approaching towards the rear of the vehicle **102**. The computer may then detect the entity **118** within the specified distance.

[0044] The vehicle computer **104** may be programmed to, upon detecting that an entity **118** is within the specified distance **122** of the vehicle **102**, actuate output of a first audio **124**. The first **124** audio may be output outside of the vehicle **102** (e.g., via a vehicle external speaker) in response to detecting the entity **118**. The first audio **124** may be any sound directed to be heard by the entity **118**. The first audio **124** to be played may be selected by the vehicle computer **104** from the stored first audio **124** based on the classification of the entity **118** as described in further detail below.

[0045] For example, the vehicle computer **104** can store and/or have access to a memory that stores various audio (e.g., sounds, words, phrases, etc.) that could be selected based on classification of an entity **118** and/or other factors. In an example, the vehicle computer **104** may store audio data (e.g., files or the like) for playback, and may further store associations between entities and/or entity classifications, whereby the computer **104**, upon identifying and/or classifying an entity **118**, can select audio data (e.g., a file) for output. This association could be stored in a table or the like. For example, the first audio **124** to be outputted to an owner of the vehicle per an audio table may be "Hello" along with the name of the owner. As another example, for specific entity classifications, an audio table could specify that no audio output is to be provided.

[0046] Example factors on which the selection of the first audio **124** by the vehicle computer **104** can include a time of day, weather conditions (e.g., presence or absence of precipitation) and/or other environmental conditions (e.g., a level of ambient light being above or below a time-of-flight threshold), a day of the week, day of the year, etc. As an example, if the vehicle computer **104** is

actuating output of the first audio **124** at 9:00 am on January 1st, the vehicle computer **104** may select the first audio **124** to be “Good morning, and happy new year.” As another example, if the vehicle computer **104** is actuating output of the first audio **124** during a storm or similarly violent weather event where a person may have difficulty hearing, the vehicle computer **104** could select the first audio **124** to be “Please seek shelter” whilst increasing the volume of the first audio **124** to compensate for any ambient noise.

[0047] As shown in FIG. 2, the entity **118** may output entity audio **126**. “Entity audio” herein means a sound made by the entity **118** and detected by vehicle audio sensors. As an example, entity **118** audio may be a query posed by, or an exclamation of distress from, an entity **118** within a specified distance **122**. Entity **118** audio is described further below.

[0048] The vehicle computer **104** may be programmed to classify the entity **118** based on entity audio **126**, camera image data, and/or any other suitable data. The classification may include any suitable technique for object classification. A vehicle computer **104** could analyze data collected about the entity **118** using a machine learning program or a rules-based program and assign a confidence score to the entity **118** (typically a percentage that indicates an estimated likelihood, e.g., 90%, 99%, etc.) that the entity classification is correct.

[0049] The classification for a specific entity **118** may be selected from a plurality of stored classifications by the vehicle computer **104**. The classifications can be determined to trigger respective actions by the vehicle computer **104** as described below. For example and without limitation, the vehicle computer **104** could classify a person detected within the specified distance **122** as threatening or nonthreatening.

[0050] The vehicle computer **104** may assign a risk score to the entity **118**. Determining a risk score may be rules-based. That is, when specified conditions are met in the process of analyzing the data, the risk score may be increased or lowered. A specified condition in the present context means any condition or phenomenon detectable by vehicle sensors **106**, and which according to programming in the computer **104** can influence the risk score of an entity **118**. Examples of specified conditions include a speed with which an entity **118** is approaching the vehicle **102**, a decibel level at which the entity **118** is outputting audio, a distance of the entity **118** from the vehicle **102** being within the specified distance **122**, etc. If a specified conditions meets a threshold, the vehicle computer **104** may reduce or increase the entity's risk score (e.g., a distance of the entity **118** below the specified distance **122** could increase the risk score, detecting audio output by an entity **118** above a decibel threshold could increase the risk score, the entity **118** moving away from the vehicle **102** could decrease the risk score, etc.).

[0051] The computer **104** could store a table or the like associating specified conditions with risk scores based on empirical testing or design considerations such as how much risk may be tolerated. Further, a detected entity **118** could initially be defined a default risk score (e.g., 50 on a scale of zero to 100, or “threatening” or “non-threatening” where the risk score is a binary determination of whether the entity **118** appears to pose a risk (i.e., is threatening) or does not appear to pose a risk (i.e., is non-threatening)). The default risk score could be a risk threshold, that is, the computer **104** could be programmed so that risk scores below the risk threshold do not trigger action, whereas risk scores at or above the threshold do. For example, the computer **104** could store data specifying risk scores for entities that cannot be identified based on a distance from the vehicle **102**. The computer **104** could alternatively or additionally store instructions to adjust a risk score based on a speed at which the entity **118** is moving with respect to the vehicle **102** (e.g., a positive speed (i.e., toward the vehicle **102**) could result in increasing the risk score, whereas a negative speed (i.e., away from the vehicle **102**) could decrease the risk score). For example, the computer **104** could detect an entity **118** within the specified distance **122**, and further could determine that the entity **118** is approaching the vehicle **102** at a speed of 5 meters per second. The computer **104** could then (e.g., according to a stored table) increase the risk score by 5 points per every 1 meter per second over 1 meter per second that the entity **118** is approaching. In this example, therefore, the entity **118**

would be assigned a risk score of 20. In this example, then the entity **118** has a risk score below the risk threshold. As another example of specified conditions, the vehicle computer **104** could detect that the entity **118** possesses a key or key fob or the like to access the vehicle **102** and therefore reduce the entity's risk score. As a further example, the vehicle computer **104** may detect, based on analyzing the data collected about the entity **118**, that the entity **118** is swinging an object. The vehicle computer **104** may then increase the risk score of the entity **118**.

[0052] Determining the risk score may further be based on stored data about the entity **118**. That is, the computer **104** may assign a risk score based on a stored risk score of an entity **118**. As an example of an adjustment of a risk score by the computer **104** based on a stored risk score, a risk score assigned to an entity **118** during a prior interaction could be retrieved from memory or obtained from a remote computer **104** upon identifying the entity **118**. In such an example, the initial risk score assigned to the entity **118** could be the retrieved risk score rather than the default risk score described above. The default risk score could continue to represent a risk threshold in such examples. In addition to the risk score being based on specified conditions being met as mentioned above, the risk score may be increased or lowered based on a stored risk score. For example, a risk score assigned to an entity **118** during a prior interaction could be retrieved from memory or obtained from a remote computer **104** upon identifying the entity **118**. In such an example, the computer **104** could assign the entity **118** a default risk score and increase or lower the risk score by an amount proportional to the stored risk score. The current risk score of the entity **118** could be increased by 5 if the stored risk score exceeded the risk threshold by 10, as an example.

[0053] The vehicle computer **104** may be programmed to assign risk scores to multiple entities **118**. That is, if more than one entity **118** is detected within the specified distance, the computer **104** may perform all the functions described throughout this document for each entity **118** individually. For example, if a first entity (not numbered) is detected within the specified distance on one side of the vehicle **102**, and a second entity (not numbered) is detected within the specified distance on a second side of the vehicle **102** at the same time, the computer may classify, assign a risk score to, identify, etc., the entities separately. The computer **104** may be further programmed to prioritize a first entity over a second entity based on the first entity being assigned a higher risk score than the second entity. That is, if two entities are detected within the specified distance and the first entity has been assigned a risk score exceeding the risk threshold, the computer **104** may treat both entities as having a risk score exceeding the risk threshold. For example, two persons may be within a specified distance of a vehicle **104**. If the first person is calm (i.e. has a low risk score) and the second person is agitated (i.e. has a high risk score) the computer **104** may treat both as having a risk score above the risk threshold and take the actions described herein. Additionally, if an entity **118** is identified by the vehicle as described below, that entity **118** may still be assigned permissions despite the presence of an entity **118** with a risk score above the risk threshold. For example, if the first entity is identified as the vehicle owner and the second entity has a risk score above the risk threshold, the first entity will not be treated as having a risk score above the risk threshold.

[0054] As described further below, the vehicle computer **104** may actuate various vehicle components **108** (e.g., an audio output component as described above) based on the risk score exceeding a threshold, for example, exceeding the default risk score. As an example, a risk score above a risk threshold would result in the vehicle **102** locking vehicle doors, closing windows, and/or sending a message to a remote device, etc. Alternatively or additionally, a risk score below a threshold could result in the computer **104** actuating one or more vehicle components **108**.

[0055] The risk score could be determined in other ways additional or alternative to initially assigning the default risk score, and may be determined based on any suitable feature or action of the entity **118**. As an example, the risk score may be based on the measured decibels of the entity **118** audio. For example, the entity **118** yelling may increase their risk score, whereas the entity **118**

speaking below a threshold decibel level may reduce, or not increase, the entity's risk score. The risk score may alternatively or additionally be determined based on physical actions being taken by the entity **118** (i.e., if the entity **118** is swinging their arms and/or holding an object) the risk score of the entity **118** may be increased. Further, the risk score may be increased or decreased based on a speed of the entity **118** measured by the sensors **106** (i.e., if the entity **118** is approaching the vehicle **102** at a specified speed or greater) this may increase their risk score.

[0056] As mentioned above, a risk score of the entity **118** may be based on stored data about the entity **118** (i.e., data that was collected or received by the vehicle computer **104** prior to a present detection of the entity **118**). The vehicle computer **104** may store data collected about the entity **118** and retrieve the stored data upon identifying the entity **118** as previously detected and one for which stored data is available. For example, an entity **118** could be identified using known recognition techniques such as image recognition or voice recognition. Typically, such recognition techniques utilize machine learning. The vehicle computer **104** may retrieve stored data for an entity **118** and compare the stored data to new or current data once the vehicle computer **104** identifies the entity **118**.

[0057] In an example, stored data about an entity **118** could be compared to currently collected data about the entity **118**. For example, the vehicle computer **104** may determine an entity's risk score based on their speaking volume and body movements as described above. This risk score and the factors on which it was based may then be stored by the vehicle computer **104**. At a later time (e.g., hours or days later) the vehicle computer **104** could then detect an entity **118** and collect further, current data about it. The stored data may then be retrieved upon a determinations that the stored data was collected from the same entity **118**, use the stored data as well as any newly collected data in making determinations relating to the entity **118**.

[0058] Additionally, the vehicle computer **104** may assign or determine permissions of an entity **118** upon identifying the entity **118** (or based on being unable to identify the entity **118**). As examples, permissions could include “permitted to operate vehicle,” “permitted to access the vehicle,” “permitted to converse with the vehicle computer **104**,” etc. Such permissions may be based on the preferences of the vehicle owner or based on determinations made by the vehicle computer **104**. For example, the vehicle owner could assign a permission to operate a vehicle **102** to another user. Further, the vehicle computer **104** may prompt the entity **118** for image data of a driver's license in order to confirm that the entity **118** is the acquaintance or could use object recognition techniques to determine the identity **118** of the acquaintance.

[0059] The entity's risk score may be increased or lowered based on data collected after outputting a first audio **124**. This data may include the tone of voice used by the entity **118**, body language of the entity **118**, whether the entity **118** is holding an object, etc. The vehicle computer **104** may further be programmed to assign permissions to the entity **118** based on the data collected after outputting the first audio **124**. As an example, the vehicle computer **104** may increase or lower the risk score of the entity **118** based on a tone of voice of the entity **118**. “Tone” as used herein refers to the inflections used by a person when speaking. Such inflections may be detected by the vehicle computer **104** and used to determine risk score and/or make classifications (i.e., if the tone of voice of the entity **118** is determined by the vehicle computer **104** to be “agitated,” the vehicle computer **104** could increase the risk score of the entity **118**). As another example, if the tone of voice of the entity **118** is determined to be “questioning,” the vehicle computer **104** may lower the risk score of the entity **118**. Tone may be measured by analyzing entity audio data for speed of speaking, volume, and pitch. For example, the computer may store a table specifying certain tones attached to certain notes. When the computer detects that the pitch of the entity audio is a certain note, the computer look up the corresponding tone for that note. The table may further store a table specifying certain tones attached to speaking speed and volume. The vehicle computer **104** may determine the tone of voice of the entity **118** using any suitable technique (e.g., a machine learning program trained to classify entities based on tones). For example, by the machine learning program,

the vehicle computer **104** may detect that a pitch of the person's voice rises at the end of a sentence which could indicate that the tone of the person is questioning, thereby lowering the risk score. As another example, the vehicle computer **104** may detect that the pitch of the person's voice remains low and gruff, which could indicate hostility and may raise their risk score.

[0060] The vehicle computer **104** may store a permissions table or the like indicating what permissions are granted to respective entities that could be identified. For example, user John Doe could be given full permission to access and operate the vehicle **102**.

[0061] The vehicle computer **104** may be programmed to classify the entity **118** based on image data from one or more vehicle sensors **106**. Classifying the entity **118** means determining a type or category of the entity **118** according to any suitable technique. For example, various techniques can be used to analyze data from cameras or other sensors **106** to determine a type or category of entity **118**. A type or category of entity **118** means a kind of object, etc., a person, a bicycle, a vehicle, a rock, etc. Image data used for entity classification and/or other purposes (e.g., determining entity behavior such as direction and/or speed of movements) may be a plurality of images collected at different times. For example, analyzing a plurality of images collected at different times may allow the vehicle computer **104** to determine if the entity **118** is walking towards the vehicle **102** or waving their hand.

[0062] An entity classification may be based on data collected about the entity **118** after outputting the first audio **124** from vehicle speakers and then detecting further data about or from the entity **118**. That is, in addition or alternative to collecting data about the entity **118** upon detecting the entity **118** within the specified distance **122** (i.e. prior to outputting the first audio **124**) the vehicle computer **104** may collect further data about the entity **118** after the vehicle computer **104** has output the first audio **124** described above. In such an example, the vehicle computer **104** could detect but not classify or identify the entity **118** prior to outputting the first audio **124**. As an example, an entity **118** could be detected within the specified distance **122**. The vehicle computer **104** may then output the first audio **124** in the form of a greeting such as "hello." The vehicle computer **104** may then collect further data about the entity **118**, and could use the further data to classify and or identify the entity **118**.

[0063] As described above the vehicle computer **104** may further be programmed to assign classifications to the entity **118** based on the data collected after outputting the first audio **124**. For example, the entity **118** may output audio as described above (i.e. the entity **118** may speak to the vehicle **102**). The entity **118** could output an audio such as asking, "Who owns this vehicle?" Then the vehicle computer **104** may collect further data about the entity **118** and classify the entity **118** as a person due to the entity's usage of language. As another example the entity **118** could output an audio such as the barking of the dog which could be used by the computer **104** to classify the entity **118** as a dog.

[0064] The vehicle computer **104** may be programmed to output a second audio based on classifying the entity **118**. The vehicle computer **104** may select the second audio from a list of sounds and phrases as described above relative to the first audio **124**. The second audio is selected by the vehicle computer **104** based on the classification of the entity **118**. For example, if the vehicle computer **104** has assigned an entity **118** a classification of "non-threatening" and given a "vehicle owner" permission (e.g. identified the entity **118** as the vehicle owner) the vehicle computer **104** may output a second audio greeting the vehicle owner by name. As another example, the second audio may be responsive to a communication by the entity **118**. If the vehicle computer **104** has assigned the entity **118** a classification of "non-threatening" and a permission of "stranger," (e.g. identified the entity **118** as a stranger) the vehicle computer **104** may output a second audio comprising an answer to a question asked by the entity **118**. In other words, the second audio may be selected from the stored plurality of phrases based on a verbal response by the entity **118** to the first audio **124**. As a further example, if the vehicle computer **104** has assigned a classification of "threatening" to the entity **118**, and the entity **118** has a risk score above a risk threshold, the

vehicle computer **104** may output a second audio requesting that the entity **118** depart.

[0065] The vehicle computer **104** may be programmed to output a second audio based on the entity audio **126**. For example, the vehicle computer **104** may have a default language. The default language may be English in the United States, German in Germany, any language selected by the vehicle owner, etc. If the entity **118** outputs an entity audio **126** in a language that is not the default language, the computer **104** may output the second audio in the language spoken by the entity **118**. The vehicle computer **104** may use known language recognition techniques to recognize the language spoken by the entity **118**.

[0066] As described above, the vehicle computer **104** may be programmed to activate a vehicle feature based on classifying the entity **118**. For example, based on classifying the entity **118** as “non-threatening” and assigning permissions of “vehicle owner,” (e.g. identifying the entity **118** as the vehicle owner) the vehicle computer **104** may unlock the doors, start the engine, or activate any feature of the vehicle **102**. Conversely, and as another example, if the entity **118** is classified as “non-threatening” and given a permission of “permitted to access vehicle,” (e.g. identified as a person known to the vehicle owner) the vehicle computer **104** may unlock the vehicle doors but not necessarily start the engine. As a further example, if the entity **118** is classified as “threatening,” (e.g. the entity **118** has a risk score above a risk threshold) the vehicle computer **104** may send a communication to the vehicle owner alerting the vehicle owner of the entity **118**.

Example Processes

[0067] FIG. **3** shows a process flow diagram of an example process **300** for detecting and interacting with entities within a specified distance **122** of a vehicle **102**. The process can be carried out according to program instructions executed by the vehicle computer **104**. The process begins in a decision block **305** in which the computer **104** determines whether an entity **118** has been detected. The detection of an entity **118** is described in greater detail above. If the computer **104** does not detect an entity **118** within the specified distance **122** of the vehicle **102**, the process continues to block **370**. If the vehicle computer **104** does detect an entity **118** within the specified distance **122** of the vehicle **102**, the process continues to block **310**.

[0068] Next, in a block **310**, the computer **104** actuates sensors **106** to collect data about the entity **118**. In some implementations, actuating or activating sensors may not be necessary because sensors may already be active. However, in some implementations, various sensors **106** such as camera sensors or the like may be inactive (e.g., to conserve power while a vehicle **102** is parked) and accordingly sensors **106** can be actuated and/activated upon an entity **118** being detected. There may be a plurality of sensors that is less than all available vehicle sensors which are powered up or active at all times to monitor for an entity **118** within the specified distance **122** as described above. For example, when a vehicle **102** is in a KEY-OFF state or the like, one or more sensors could be provided to monitor vehicle surroundings (e.g., to provide a field or fields of view **120** around the vehicle **102**) and detect entities. Once an entity **118** is detected, the vehicle computer **104** could then actuate the remaining sensors **106** to begin collecting further data about the entity **118**.

[0069] Next, in a block **315**, the computer **104** compares the data collected in block **310** to stored data. As described above, the computer **104** may store data collected about an entity **118** to be retrieved later. The comparison of stored data to the newly collected data may begin once collection of data has begun.

[0070] Next, in a block **320**, the computer **104** classifies the entity **118**. As described above, classification of the entity **118** may be based on data collected from the entity **118** and on previously stored data about the entity **118**.

[0071] Next, in a decision block **325**, the computer **104** determines whether to output a first audio **124**. The computer **104** may determine not to output a first audio **124**. For example, if the entity **118** is classified as an animal or as another vehicle, the computer **104** may not output a first audio **124**. Likewise, if the entity **118** is a person that passes within the specified distance **122** and moves away (i.e. walks by the vehicle **102**) the computer **104** may not output a first audio **124**.

[0072] Next, in a block **330**, the computer **104** outputs first audio **124** such as described above. The vehicle computer **104** may select the first audio **124** to be played from the stored first audios based on the classification and identification of the entity **118**. As an example, if the entity **118** is identified by the computer **104** as a stranger asking a question, the first audio **124** may be an answer to the question. If the entity **118** is identified as the owner of the vehicle **102**, the first audio **124** may be a welcome to the owner.

[0073] Next, in a block **335**, the computer **104** stores the data collected about the entity **118**. As described above, the data can be later retrieved to identify and/or assign a risk score to the entity **118**.

[0074] Next, in decision block **340**, the computer **104** determines whether the entity **118** has been identified. That is, an individual entity **118** (e.g., a person) can be identified using a suitable technique as described above. As an example, if the vehicle computer **104** detects an audio output of the entity **118**, the vehicle computer **104** may use a voice recognition technique to compare the audio to stored audio data and then identify the entity **118**. If the stored audio data and the newly collected audio data are determined by the computer **104** to be from the same entity **118**, then the computer **104** tags the entity **118** as identified. If the entity **118** is identified the process continues to block **345**. Otherwise, the process continues to block **350**.

[0075] In a block **345**, the vehicle computer **104** has identified the entity **118**. The vehicle computer **104** then could grant vehicle permissions to the entity **118**. Permissions may be based on a permissions table or the like as described above. Further, the entity **118** could be identified but have no associated permissions, in which case none may be granted.

[0076] In decision block **350**, the computer **104** has previously determined that the entity **118** is not identified by the vehicle computer **104**. The computer **104** then analyzes the data collected in block **310** for the specified conditions to determine the risk score of the entity **118** as described above. If the computer **104** increases the entity's risk score to the risk threshold or beyond the risk threshold, the process continues to block **355**. Else, the process continues to decision block **360**.

[0077] Next, in a block, **355** the vehicle computer **104** can send a message to a remote device. The remote device may be any suitable computing device other than the vehicle computer **104**, such as a mobile device or smartphone or the like. When the specified conditions are detected in the previous block (i.e. the entity's risk score exceeds the risk threshold) the vehicle computer **104** may send a message to the mobile device of the vehicle owner alerting them of the entity **118** and/or send a message to a law enforcement device requesting assistance.

[0078] Next, in a decision block **360**, the computer **104** determines whether a second output should be output. The computer **104** may determine not to output a second audio. For example, if the entity **118** was identified as a person asking a question and the first audio **124** answered the question thus causing the entity **118** to depart, the computer **104** may determine not to output a second audio. As another example, if the entity **118** was identified as the vehicle owner and the vehicle owner has entered the vehicle **102**, the computer **104** may determine not to output a second audio. As described above, the vehicle computer **104** may output the second audio in a language of the entity **118** (i.e. Spanish if the entity audio was Spanish). As a further example, if the entity **118** has been assigned a risk score above the risk threshold, the computer **104** may output a second audio as mentioned below. If the computer **104** determines to output a second audio, the process continues to block **365**. Otherwise, the process continues to decision block **370**.

[0079] Next, in a block **365**, the computer **104** actuates a vehicle component **108** to output a second audio. As described above the vehicle computer **104** may select the second audio from a list of sounds and phrases as described above relative to the first audio **124**. The second audio is selected by the vehicle computer **104** based on the risk score and/or identification of the entity **118**. That is, the second audio is selected to conform to the risk score and/or of the entity **118**. For example, if the vehicle computer **104** has identified the entity **118** as the owner of the vehicle **102**, the vehicle computer **104** may output a second audio greeting the vehicle owner by name. As another example,

the second audio may be responsive to a communication by the entity **118**. If the vehicle **102** has identified the entity **118** as a person asking a question, the vehicle computer **104** may output a second audio comprising an answer to the question asked by the entity **118**. In other words, the second audio may be selected from the stored plurality of phrases based on a verbal response by the entity **118** to the first audio **124**. As a further example, if the vehicle computer **104** has assigned a risk score to the entity **118** above a risk threshold as described above, the vehicle computer **104** may output a second audio requesting that the entity **118** depart.

[0080] In decision block **370**, the computer **104** determines whether to continue the process. For example, once the process **300** is initiated, the computer **104** could continue to monitor for the detection of entities **118** within the specified distance **122** of the vehicle **102** by returning to block **305**. However, the process could end upon some input or event to terminate the process, such as a vehicle user commencing operation of the vehicle **102** (e.g., activating a propulsion system such as an engine) a user providing input to end the process, etc. If the process **300** is to continue, then the process returns to the block **305**, otherwise the process **300** ends.

[0081] Computing devices such as those discussed herein generally each includes commands executable by one or more computing devices such as those identified above, and for carrying out blocks or steps of processes described above. For example, process blocks discussed above may be embodied as computer executable commands.

[0082] Computer executable commands may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Python, Julia, SCALA, Visual Basic, Java Script, Perl, HTML, etc. In general, a processor (i.e., a microprocessor) receives commands (i.e. from a memory, a computer readable medium, etc.) and executes these commands, thereby performing one or more processes, including one or more of the processes described herein. Such commands and other data may be stored in files and transmitted using a variety of computer readable media. A file in a computing device is generally a collection of data stored on a computer readable medium, such as a storage medium, a random access memory, etc.

[0083] A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (i.e., tangible) medium that participates in providing data (i.e., instructions) that may be read by a computer **104** (i.e., by a processor of a computer **104**). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Instructions may be transmitted by one or more transmission media, including fiber optics, wires, wireless communication, including the internals that comprise a system bus coupled to a processor of a computer **104**. Common forms of computer-readable media include, for example, RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer **104** can read.

[0084] All terms used in the claims are intended to be given their plain and ordinary meanings as understood by those skilled in the art unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as “a,” “the,” “said,” etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary.

[0085] In the drawings, the same candidate numbers indicate the same elements. Further, some or all of these elements could be changed. With regard to the media, processes, systems, methods, etc. described herein, it should be understood that, although the steps or blocks of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. In other words, the descriptions of processes herein are provided for the purpose of illustrating certain embodiments, and should in no way be construed so as to limit the claimed invention.

[0086] Use of in response to, based on, and upon determining herein indicates a causal relationship,

not merely a temporal relationship. “Based on” or “in response to” can mean based at least partly on or at least partly in response to unless explicitly stated otherwise.

[0087] Examples are contemplated herein. Any example embodiment or feature described herein is not necessarily to be construed as preferred or advantageous over other embodiments or features. Further, the example embodiments described herein are not meant to be limiting. It will be readily understood that certain aspects of the disclosed systems and methods can be arranged and combined in a wide variety of different configurations, all of which are contemplated herein. In addition, the particular arrangements shown in the figures should not be viewed as limiting. It should be understood that other embodiments might include more or less of each element shown in a given figure. Additionally, some of the illustrated elements may be combined or omitted. Yet further, an example embodiment may include elements that are not illustrated in the figures.

[0088] The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described. It will be understood that the usage of the terms “first” and “second” are merely identifying and not necessarily indicative of priority.

Claims

1. A system, comprising a computer having a processor and a memory, the memory storing instructions executable by the processor to: detect an entity within a specified distance of a vehicle; output a first audio outside of the vehicle in response to detecting the entity; classify the entity based on data collected about the entity after outputting the first audio; and output a second audio based on classifying the entity.
2. The system of claim 1, the instructions including further instructions to activate a vehicle feature based on classifying the entity.
3. The system of claim 1, wherein the detection of the entity within the specified distance of the vehicle is based on image data.
4. The system of claim 1, wherein the second audio is responsive to a communication by the entity.
5. The system of claim 1, wherein the second audio is selected from a stored plurality of phrases.
6. The system of claim 1, the instructions including further instructions to identify the entity based on stored data collected from the entity prior to outputting the first audio.
7. The system of claim 1, the instructions including further instructions to classify the entity based on a plurality of images collected at different times.
8. The system of claim 1, the instructions including further instructions to classify the entity based on a tone of the entity.
9. The system of claim 1, wherein the classification of the entity is at least one of threatening and nonthreatening.
10. The system of claim 1, wherein the first audio is output in a first language and the second audio is output in a second language based on detecting the second language in an entity audio.
11. The system of claim 1, further comprising detecting a first entity and a second entity within the specified distance of the vehicle, the first entity being prioritized based on a risk score.
12. A method comprising: detecting an entity within a specified distance of a vehicle; outputting a first audio outside of the vehicle in response to detecting the entity; classifying the entity based on data collected about the entity after outputting the first audio; and outputting a second audio based on classifying the entity.
13. The method of claim 12, further comprising activating a vehicle feature based on classifying the entity.
14. The method of claim 12, wherein detecting the entity within the specified distance of the

vehicle is based on image data.

15. The method of claim 12, wherein the second audio is responsive to a communication by the entity.

16. The method of claim 12, wherein the second audio is selected from a stored plurality of phrases.

17. The method of claim 12, further comprising identifying the entity based on stored data collected from the entity prior to outputting the first audio.

18. The method of claim 12, further comprising classifying the entity based on a tone of the entity.

19. The method of claim 12, wherein the classification of the entity is at least one of threatening and nonthreatening.

20. The method of claim 12, wherein the first audio is output in a first language and the second audio is output in a second language based on detecting the second language in an entity audio.
